Pipelines & Tanks

Farm-level water and fertilizer plumbing — one shared tank and one shared fertigation rig per farm, distributed out to every greenhouse room's own misting manifold.

Scope	Kadawatha Farm (3 greenhouse rooms, 5 benches each) · Raddolugama Farm (2 rooms, 5 benches each)
Architecture	1 water tank + 1 fertigation rig per farm · 1 environment sensor + misting per room · benches carry no plumbing of their own
Controllers	Farm controller (tank + rig) and one controller per room (sensor + misting) — see backend/src/routes/device.js
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This system does not give each bench its own tank or sensor. Physically, a farm has one water tank and one fertilizer mixing/dosing rig, plumbed out to however many greenhouse rooms sit on that site; each room has its own single environment sensor and runs misting for every bench in it together. The tables and diagram below describe that shared plumbing — not per-bench hardware. For electrical wiring and GPIO pin assignments, see `hardware-and-wiring-reference.pdf` in this same folder.

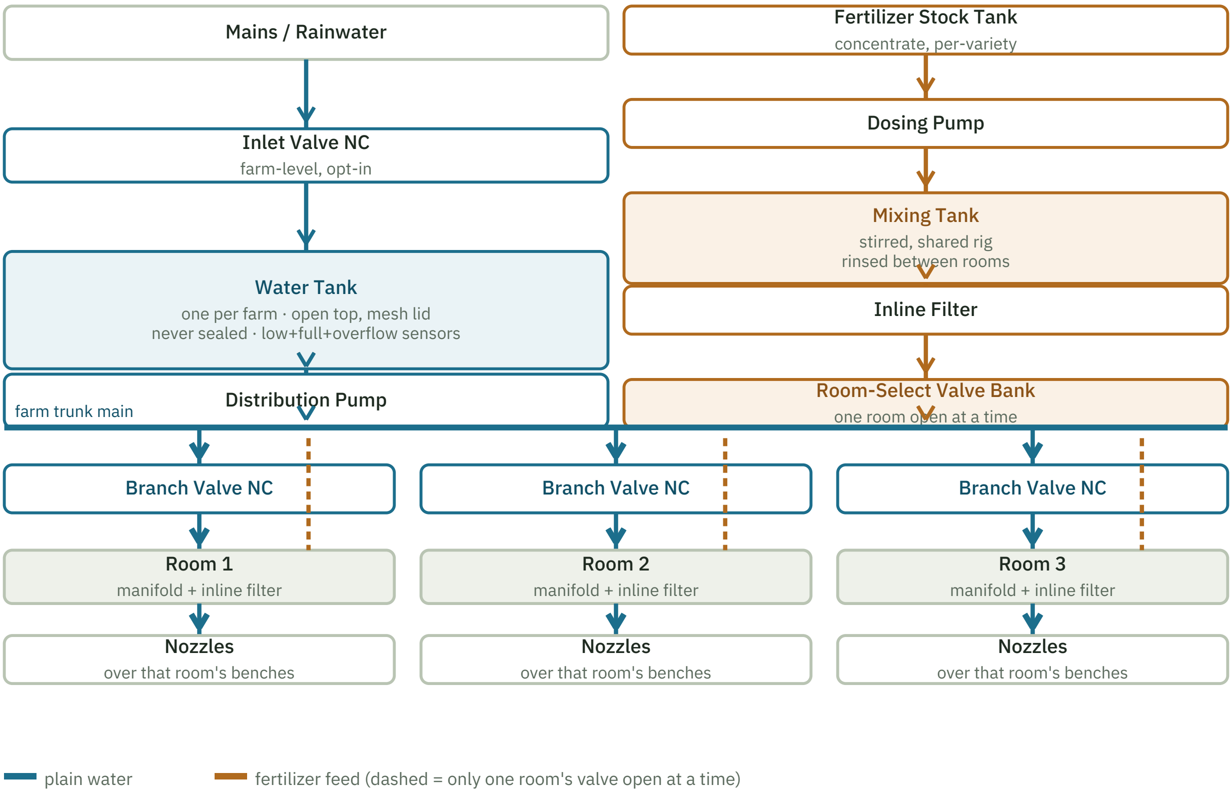
Tank Inventory

Per farm, not per room or bench.

Tank	Owner	Contents	Notes
Water Tank	1 per farm	Mains/rainwater, chlorinated	Open top, loose mesh lid only — never sealed, so chlorine can off-gas. Low + full + overflow sensors clamp to the outside wall. Feeds every room's misting through the trunk main.
Fertilizer Stock Tank	1 per farm	Concentrated fertilizer, hand-mixed	Source for the dosing pump. Early- and late-phase products (per the active room's rotation week) are swapped in here between cycles — the rig doesn't hold two products at once.
Mixing Tank	1 per farm (shared rig)	Concentrate + batch water, stirred	Rinsed between rooms. Each room's schedule sets its own mix ratio (mL/L) and batch volume, so back-to-back feeds for different rooms can use different strengths.
Fungicide Tank	1 per farm (optional, manual by default)	Wettable-powder fungicide, hand-mixed	Entirely separate plumbing run, start to finish — never joins the water or fertilizer manifold. Sprayed through its own dedicated coarse nozzle so powder residue can't clog the fine fogging nozzles.

Farm Distribution Diagram

Blue = plain water, amber = fertilizer feed. The room-select valve bank is the physical counterpart of the software's fertigation lock (see backend/src/routes/device.js, `fert-status`) — only one room's feed valve is ever open at a time, because there is only one mixing tank.



Room Branch — What Each Greenhouse Has Locally

Component	Detail
Branch shutoff valve (water)	Normally-closed, on the trunk main. That room's controller opens it for the duration of a mist cycle, same as the old per-bench design — just now gating the whole room at once.
Local manifold	Where the room's water branch (and, only while it's that room's turn, the fertilizer feed) reaches every bench's nozzle line.
Inline filter	Right before the nozzles — last line of defense against clogging, per room.
Nozzles	Fine fogging nozzles (30 µm class) over that room's benches. No fertilizer/fungicide line ever shares this run with anything but the water branch and, briefly, the shared feed line.
No local tank	A room holds no water or concentrate of its own — it draws entirely from the farm's shared tank and rig on demand. If the farm tank is low or dechlorinating, every room under it is held back together.

Valve & Routing Table

Connection	Detail
Mains/Rainwater → Inlet Valve → Water Tank	Opens when the low sensor clears, closes on the full sensor or the safety timeout. Farm-level, one inlet for the whole site.
Water Tank → Distribution Pump → Trunk Main	Single straight run, no branching until the trunk. Strainer/mesh at the tank outlet.
Trunk Main → Room Branch Valve (×N)	One normally-closed valve per room, tee'd off the trunk. Only the requesting room's valve opens; the others stay shut, so one room misting doesn't drain pressure meant for another.
Room Branch Valve → Room Manifold → Nozzles	The daily plain-misting path, local to that room.
Fertilizer Stock Tank → Dosing Pump → Mixing Tank (stirred)	Concentrate metered into batch water at that room's configured ratio; the tank is rinsed before the next room's cycle if the product differs.
Mixing Tank → Inline Filter → Room-Select Valve Bank	One shared filter, then a bank of normally-closed valves — one per room, interlocked so at most one is open.
Room-Select Valve → check valve → Room Manifold	The check valve is non-negotiable on every room's feed line — without it, fertilizer can siphon back into the shared trunk or another room's branch.
Fungicide Tank → Fungicide Pump → Fungicide Valve → dedicated coarse nozzle	Its own plumbing run per farm (or per room, if sprayed locally) — never joins the water/feed manifold at any point.
Water level sensors (low / full / overflow)	All three clamp to the outside of the one farm tank wall — non-contact, at three heights.

Sizing & Capacity

Reference sizing for the current fleet shape — scale the trunk and tank up before adding more rooms than this, not after.

Farm	Rooms × benches	Suggested tank size	Trunk / branch pipe	Distribution pump
Kadawatha Farm	3 × 5 = 15 benches	≥100L water, 30–50L fertilizer stock	¾" trunk, ½" branch per room	≥4.5 L/min, sized for 1 room misting at a time
Raddolugama Farm	2 × 5 = 10 benches	≥75L water, 30L fertilizer stock	¾" trunk, ½" branch per room	≥4.5 L/min

The 50L single-bench barrel from the original test-bench build (`hardware-and-wiring-reference.pdf`) does not carry over once a farm has more than one room — that was sized for 10 pots on one manifold, not a multi-room trunk line.

Interlocks & Safety

- One room fed at a time. There is one mixing tank per farm, so the room-select valve bank is hardware-interlocked (or at minimum firmware-interlocked at the farm controller) to hold every valve shut except the one room currently being dosed — matches the software's `withFarmLock` serialization in the simulator/backend.
- Dechlorination hold applies to the whole farm at once. Because every room shares one tank, a fill event holds back misting for every room under that farm for the configured hold (24h default), not just whichever room triggered the fill.
- Check valves on every fertilizer branch. Non-negotiable on each room's feed line, not just the trunk — a stuck-open branch valve elsewhere must not let concentrate migrate into a room that isn't being fed.
- Fungicide never touches the fine manifold. Same rule as the single-bench build, now repeated at every room: its own tank, pump, valve, and dedicated coarse nozzle, start to finish.
- Overflow cutoff is farm-wide. One mechanical float switch above the full sensor on the shared tank protects every room at once — there's no per-room backup.

Before You Plumb It

1. Confirm the room-select valve bank defaults to all closed on power-loss — a fail-open valve here means two rooms could receive fertilizer at once, or one room drains the mixing tank meant for another.
2. Pressure-test the trunk main before tying in room branches — a leak upstream of the branch valves affects every room, not just one.
3. Physically trace the fungicide run end-to-end and confirm it never tees into the water/feed manifold at any room, not just the first one built.
4. Size the trunk main for the worst case of one room's full mist + feed draw, not the sum of all rooms — the interlocks mean only one room draws from the shared rig at once, and misting across rooms is not synchronized to overlap by design (though it isn't prevented, since plain water has no cross-room interlock).
5. Don't seal the water tank lid — same rule as the single-bench build, now protecting every room's supply at once.

6. Label the room-select valve bank physically (Room 1 / Room 2 / Room 3...) matching the dashboard's room names — the software's room IDs mean nothing to someone standing at the manifold with a wrench.